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notes on papers

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(40) annotations: Belyaev 2023b(1).pdf

paper: *Table 5 from SMOR* (n.d.)

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note

comment

2

from the point of view of context-free rules, VP and V are atomic symbols that are not related to each other; labeling one of the daughters of NP as N is merely a convention

2

Briefly, c-structure is a phrase structure tree; constraints on possible trees are usually described via context-free rules as in

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) it has become a universal practice to capture long-distance dependencies through functional uncertainty at fstructure, and the use of empty categories at c-structure has become unnecessary

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enforces endocentricity by introducing the notion of projection and “bar level” and requiring that each non-maximal projection (X_0 and X' ; X'' , or XP , is usually assumed to be the maximum level of projection) be dominated by a node belonging to the same category, with the bar level either incremented by one or unchanged. The sisters of c-structure heads (complements, specifiers and adjuncts) have to be maximal projections or non-projecting words (on which see below).

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wrong with this structure: an I head cannot be the daughter of NP; the VP cannot be headed by, or even immediately dominate, a Det; an AdvP cannot be headed by a noun.¹ The principle that prohibits this is called endocentricity; roughly stated, it means that the external distribution of a phrase (e.g. NP) is determined by the category of one and only one of its daughters, the head.

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Second, X'-theory in LFG allows for the following positions: complement (3a), specifier (3b), X'-adjunct (3c) and XP adjunct (3d).

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Specifically, heads may only be stipulated if there is actual lexical material that can occupy them; therefore, even the existence of projections such as CP or IP cannot be automatically assumed for all languages. More abstract projections such as TopicP or ForceP are not usually introduced because there are few suitable candidates for the status of heads of these phrases, and little distributional evidence to argue that their specifiers are distinct structural positions

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most prominent exception is the exocentric category S.³ This category does not have a "head" in the normal sense: it can be "headed" by a verb, but also by an adjective or another nonverbal predicate; this is why the term S is used instead of, for example, VP. The category S is most extensively used in nonconfigurational languages

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lexicalism implies that the features of individual syntactic elements (morphemes and word-forms) as well as their subcategorization frames are determined in the lexicon, and cannot be modified in the syntax (such as by promoting the direct object in a passive construction)

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as a result, X'-theory, understood purely in terms of c-structure, is little else than a system of labeling nodes which allows us to generalize endocentricity at constituent structure level.

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syntax can only multiply define lexical features, but cannot override them.

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individual words that are constructed from different blocks and according to different rules than syntactic constituents.⁵ Thus, the distinction between morphology and syntax in LFG is viewed as fundamental

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This understanding of lexicalism is more formally termed lexical integrity

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since all morphology is sublexical, the position of individual affixes cannot have any syntactic relevance

9

there is no such thing as two different f-structures having the same set of attributes and values;

9

NA

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9

, c-structure in LFG is complemented by an additional level of representation called f-structure. F-structure is an attribute-value structure that includes information on valency, grammatical functions, and the features of clauses and their syntactic arguments.

9

F-structure is usually thought of as a set of attribute-value pairs, or a function that maps attribute names to their values

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all bundles of agreement features (agr) with the same set of values are identical to each other. One agr bundle may be required to be identical to another via agreement sharing (Haug & Nikitina 2015) in the f-description (using an equation such as $(\square \text{ agr}) = (\square \text{ subj agr})$), but it will also be identical to all other such bundles elsewhere in the same sentence, if they occur.

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uniqueness of pred values (Section 3.3.4), which ensures that any two independently introduced, semantically interpreted f-structures are formally distinct, even if they have the same lexical predicate and the same set of morphosyntactic features

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default relation between equations forming an f-description is conjunction, but disjunction is also possible; for example, $\{ (\square \text{ subj pers}) = 1 \text{ } (..) (\square \text{ subj pers}) = 2 \}$ means that the subject is defined as being either 1st or 2nd person

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a value cannot be assigned by a constraining equation

11

they are “constructive”, or defining: informally, a defining equation introduces a feature value, regardless of whether it is the only such equation or the same value is defined elsewhere

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constraining equations serve as a good illustration of the LFG principle of separation between description and the object being described

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The only thing a constraining equation does is to put constraints on permissible structures; it does not contribute to the structures themselves.

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The other two types of constraining equations are existential and negative constraints. Existential equations check that a feature has any value rather than testing for a specific value. They are written as simple function applications: (ϕ ψ) means that the f-structure ψ must have the feature ϕ with any value; the absence of an equality statement indicates that we are dealing with an existential constraint. Negative constraints check that a feature does not have a given value ($(\phi \psi) \neq \psi$; this is compatible with the feature having

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resolution of a valid f-structure for a sentence must proceed through two steps: (a) evaluation of defining equations; (b) evaluation of constraining equations

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island constraints

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In the interaction between c and f-structure, c-structure is exclusively concerned with linear order and hierarchical embedding, while f-structures do not reflect linear order or constituent structure in any way. Therefore, linear order is relevant for most morphosyntactic constraints only in a limited way, insofar as it distinguishes between different c- to f-structure mappings

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agreement features are the domain of f-structure, and functional equations can only refer to f-structure functions, not linear or constituent-based positions.

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In LFG, such constraints can be captured using the relation of f-precedence (Kaplan & Zaenen 1989a), which is a way of introducing linear order constraints in f-structure using the inverse projection ϕ^{-1} , which maps f-structures to the corresponding c-structure nodes

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there is nothing in the formal architecture or in any part of an LFG grammar that would prohibit a “clausal” f-structure to have the feature case or a “nominal” f-structure to have the feature tense; such constraints are only implicit in the way these f-structures are constructed and mapped from c-structure nodes

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semantic form is a special type of attribute value that is exclusively assigned to the attribute pred. Semantic forms consist of the predicate name followed by the list of its syntactic arguments

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16fn is an abbreviation for functor; sfid stands for “semantic form identifier

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There are three conditions that any f-structure must satisfy in order to be treated as valid: Uniqueness (also known as Consistency), Completeness, and Coherence.(..)(..)Any f-structure that violates these conditions cannot be part of a valid analysis of any sentence, regardless of the rules of a particular grammar.

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36) Ill-formed f-structure for *John, Mary saw John

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37) Ill-formed f-structure for *Mary saw:

#q: veni, vidi, vici ? how w/o context pred ‘videre<subj.empty>’ ? </subj.empty>

31

Glue Semantics is resource-sensitive, which automatically ensures both Completeness and Coherence: Completeness, because all premises of the meaning constructor introducing the main predicate have to be saturated; Coherence, because no unused resources have to be left

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Such MWEs often span whole syntactic phrases, and the lexical constraints involved cannot be captured locally. One solution that has been proposed in LFG is to replace the contextfree c-structure by a variant of Tree-Adjoining Grammar

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the larger the tree, the longer its f-description; due to monotonicity, the f-structure of a larger tree fragment is, then, always more specific than the f-structure of any of its subtrees

(7) annotations: Boerjars 2020 LFG Annual Review 6_1.pdf

paper: Belyaev (2023)

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note

comment

2

crucial idea underpinning Lexical-Functional Grammar (LFG) is that a linguistic element of any size—a word, a phrase, or a sentence—is associated with different types of linguistic information, for instance, information about prosody, about category and constituency, about grammatical relations, and about semantics

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It is a parallel correspondence model because the different dimensions of linguistic information are represented separately but are connected by well-defined principles of correspondence, or mappings.

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“the formal model of LFG is not a syntactic theory in the linguistic sense. Rather, it is an architecture for syntactic theory

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c-structure (constituent and category)

3

C-structure captures information about constituency and categories

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Oblique argument

prepositional argument, obligatoire gehen1: nach, in > oblig. gehen2: walken, laufen, gehen (mit stöckern): nicht oblig.

6

Adjunct

not necessary specification, vom Prädikat nicht verlangt

Belyaev, Oleg. 2023. “Introduction to LFG.” *Handbook of Lexical Functional Grammar* (Berlin), November, 3–22. <https://doi.org/10.5281/zenodo.10185934>.

Table 5 from SMOR: A German Computational Morphology Covering Derivation, Composition and Inflection | Semantic Scholar. n.d. Accessed May 13, 2026.
<https://www.semanticscholar.org/paper/SMOR%3A-A-German-Computational-Morphology-Covering-Schmid-Fitschen/9e46cd82175bdacfd78546efea07ac5b018e0ace>.